

A multi-agent system for automating scientific discovery

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Referee #1

(Remarks to the Author)

The manuscript by Ghareeb et al. presents Robin, a multi-agent system designed to automate the scientific discovery process by integrating literature synthesis with experimental data analysis. The system architecture consists of several specialized agents: Crow and Falcon for literature review and Finch for autonomous analysis of experimental data, such as flow cytometry and RNA-seq. The core workflow involves Robin identifying a therapeutic strategy for a given disease, proposing testable drug candidates, delegating experiments to human scientists, and then analyzing the resulting data to refine its hypotheses in an iterative "lab-in-the-loop" framework.

The system's capabilities are demonstrated through a real-world use case: identifying a novel treatment for dry age-related macular degeneration (dAMD). Starting from the broad disease prompt, Robin identified impaired retinal pigment epithelium (RPE) phagocytosis as a key mechanism. After reviewing approximately 400 publications, it nominated 30 repurposable drugs. Experimental validation of the top candidates confirmed that the ROCK inhibitor Y-27632 enhanced phagocytosis. In a subsequent iteration informed by these results, Robin proposed ripasudil—a ROCK inhibitor approved in Japan—which was then experimentally shown to be even more effective. Finch, the data analysis agent, autonomously analyzed the raw flow cytometry and RNA-seq data, identifying the upregulation of ABCA1 as a potential underlying mechanism.

Overall, this work represents one of the first published examples of an AI system driving a complete discovery cycle, from hypothesis to an experimentally validated therapeutic candidate.

Strengths

1. The system's end-to-end architecture moves beyond "idea generators" by creating a closed-loop system where the data analysis agent, Finch, writes and executes Jupyter notebooks to inform the next round of hypothesis generation.
2. A successful lab-in-the-loop demonstration validates the approach by closing the loop with original flow cytometry and RNA-seq data and showing that both autonomous and human analyses reach the same qualitative conclusion.
3. The system's focus on drug repurposing demonstrates a pragmatic approach. Identifying ripasudil, a drug with an established clinical safety profile, highlights how this method could accelerate development timelines and address the clear unmet need for dAMD treatments.
4. The workflow is presented with sufficient detail, as the authors provide information on the prompts, agent architecture, and ranking schemes used, which is crucial for allowing others to replicate or critique the pipeline.
5. The use of LLM-based judging to infer a global ranking from many individual pairwise contests is a useful mechanism for automating decision-making. Its implementation here is strengthened by benchmarking against human experts, which shows strong concordance.

Weaknesses and opportunities for improvement:

1. The claim of being the "first system to fully automate..." is overstated. Several sophisticated multi-agent frameworks have been described recently, including Google's "AI Co-Scientist," Boiko et al.'s "Coscientist" (Nature, 2023), and "TxGemma" (Wang et al., 2025, arXiv). Robin's specific contribution is its focus on biological discovery and its integration of an autonomous data analysis agent (Finch). The claim of primacy should be narrowed to this specific achievement.
2. The novelty of ROCK inhibition is clinically actionable but mechanistically incremental. While repurposing an existing

drug is a valuable outcome, the paper would benefit from framing its contribution carefully. The core finding is not the discovery of a new mechanism but rather the identification of a new therapeutic application for a known pathway, given that Y-27632 was previously shown to restore phagocytic efficiency (Mao & Finnemann, 2021).

3. The reported magnitude of the ripasudil effect requires reconciliation. The manuscript should address the discrepancy between the AI's reported 7.5-fold increase in MFI and the human analysis showing a 1.75-fold increase, as this points to potential instability in the autonomous analysis pipeline.

4. The paper lacks benchmarking and ablation studies, which are needed to rigorously validate the architecture. The authors should provide: (a) A comparison against a single-agent baseline (e.g., a state-of-the-art model like o3/Gemini 2.5 pro/Claude 4 given the same high-level prompt) (b) Ablation studies that quantify the marginal contribution of each agent (Crow, Falcon, Finch). (c) A head-to-head comparison on an open benchmark eg GPQA to contextualize Robin's performance against other systems. (d) A runtime and cost analysis (e.g., GPU hours or dollar cost per iteration) to assess scalability.

5. The biological validation would benefit from more translationally relevant models. The work relies solely on immortalized ARPE-19 cells. Because dry-AMD pathogenesis is still poorly understood, the proposed mechanism may ultimately fall short in a clinical setting. Including validation in primary RPE cells, retinal organoids, or an in-vivo model would strengthen the findings.

6. The generalization claims are illustrative rather than validated. The paper lists 10 further diseases with candidate tables, but provides no experimental follow-up, making them unproven examples of the system's ideation capability.

7. The statistical reporting should be more detailed. To clarify the experimental findings, the authors should report the number of biological and technical replicates for each assay, include raw effect sizes, and provide adjusted P-values.

8. The reproducibility of the Finch agent is a concern, and its stochastic nature would benefit from further characterization with run-to-run variance metrics.

9. The human effort involved is not quantified. To substantiate claims of automation, the authors should provide a time-on-task table detailing the minutes of scientist involvement required for each stage of the discovery loop.

10. The data hand-off between agents is opaque. The manuscript does not provide the precise, machine-readable artifact that passes Finch's analysis to Crow/Falcon. Providing a concrete example of that prompt or data structure is necessary to assess the degree of automation versus hidden human curation.

11. A literature quality-control filter is needed. To mitigate the "garbage-in, garbage-out" risk, the system would be improved by having the Crow agent assign a confidence score to sources (e.g., based on citation count or journal impact) before they are consumed by other agents.

12. The manuscript should ideally include a discussion of ethics and safety, noting that automated discovery pipelines should incorporate guardrails to address potential dual-use concerns or off-target effects.

Reviewer: Olivier Elemento

(Remarks on code availability)

The authors provide a GitHub repository for the Robin system. After getting a FutureHouse API key, resolving several Python version and package dependency issues (including for the aviary framework), I was able to run the demonstration notebook. The system successfully executed the initial literature search and assay generation steps. However, the workflow failed during the subsequent detailed investigation step due to a 429 Too Many Requests API rate-limiting error from the backend service. This suggests that while the code is available, successfully reproducing the full discovery loop may require direct support from the authors or modifications to handle API limits. I also note that the Finch agent is not available via github and is only accessible upon request to FutureHouse.

Referee #2

(Remarks to the Author)

The authors have developed Robin, an AI system to develop and test hypotheses with the goal of accelerating drug development. Designed to emulate the iterative scientific method, it integrates hypothesis generation, literature synthesis, and data analysis into a single workflow. In this manuscript they demonstrate Robin's capability by analyzing existing data and nominating a candidate for dry AMD, Ripasudil a ROCK inhibitor. The authors claim novelty, but how novel is questionable. It is known that phagocytosis is decreased in AMD, it is known that ROCK inhibitors increase RPE phagocytosis, it is known that Ripasudil can stimulate phagocytosis, albeit in a different cell type. Stimulating RPE phagocytosis has been contemplated for AMD (although not as a promising approach). Ripasudil has been tested in RPE models to see if it inhibits RPE inflammation that can contribute to AMD. ROCK inhibitors including Ripasudil are used in ophthalmology practice. Hence, for any expert in the field, these ideas are not particularly novel. Furthermore, the wet lab validation experiments are poorly designed and the data are not robust. In addition, it is most disconcerting that several references were mis-cited or misquoted in an effort to bolster their viewpoints. This undermines trust in the entire MS.

1. Novelty claim: Abstract: -Statement that "we were able to identify a novel treatment for dry age-related macular degeneration (dAMD), the major cause of blindness in the developed world." This is far overstated. They have not discovered a treatment for dry AMD and should be edited. Similarly, the statement "Ripasudil is a clinically-used rho kinase (ROCK) inhibitor that has never previously been proposed for treating dAMD."

While this may be the case for this specific ROCK inhibitor, the concept that ROCK inhibition might benefit dry AMD is in the literature, as examples:

PMID: 40250716 “Inflammatory processes within the choriocapillaris and retinal pigment epithelium (RPE) cell death are also thought to be early features of AMD, which ROCK inhibitors have the potential to combat.”

PMID: 27917311 Croze, et al 2016 “A closer examination of ROCK inhibition in wound healing could prompt a novel therapeutic for GA, a devastating progression of AMD.”

PMID: 39594655 “Consequently, AMD, a neurodegenerative disease characterized by the accumulation of lipofuscin, may benefit from the upregulation of autophagy through ROCK inhibition.”

PMID: 32252692 – Ripasudil was tested to see if it reduced RPE inflammation which can contribute to AMD, with positive results.

2. Novelty claim “Notably, while ROCK inhibitors have been previously suggested for treatment of wet AMD and other retinal diseases of neovascularization, Robin is the first to propose their application in dry AMD for their effect on phagocytosis [35]. This approach is supported by several lines of evidence, as RPE phagocytic dysfunction is observed not only with normal aging but is pronounced in AMD patients [41, 42, 43].”

35) ROCK inhibitors in ophthalmology: A critical review of the existing clinical evidence

42) PMID: 37371481. This paper considers stimulating RPE phagocytosis and autophagy in dry AMD treatments and explains why targeting phagocytosis is likely not the best route. This demonstrates that increasing phagocytosis as an AMD treatment has been considered in the literature.

PMID: 33024620. Ripasudil increased phagocytosis in trabecular meshwork cells in pig eyes.

Authors state “Preclinical models have demonstrated that Y-27632 can restore phagocytic efficiency in RPE cells [33], confirming Robin’s literature-based rationale for suggesting this candidate.

I checked reference 33, there is no mention of Y-27632.

3. Novelty claim: “In investigating the basis for Robin’s initial identification of Y-27632, we found that the literature search included a single paper demonstrating Y-27632’s ability to enhance phagocytic efficiency in “low-phagocytic” RPE cells from human donors by promoting actin polymerization [42]. “

42 – I looked at that citation and could not find any reference to Y-27632. Hence, again, this is a mis-cited reference.

However this is described in PMID: 34440696, and ROCK inhibition stimulating RPE phagocytosis is also described in PMID: 29456184 and PMID: 40389756. Hence, there are at least 3 relevant papers that the authors should have cited.

Therefore the authors claim the novelty of their findings by not fully describing the literature.

4. Validation using RPE cells. The authors should also have followed Robin’s direction to use PSC-RPE as well as ARPE19 cells. Instead, they just used ARPE19 cells, which are not a good model of authentic RPE, having a different transcriptome, morphology, and phagocytosis. Moreover, there is considerable variation in the line, depending on the source and how it is grown.

This paper provides guidance on what should be provided in publications using ARPE19 cells: PMID: 35314139.

They should submit the information requested in that publication. For example, they claim that the cells show signs of differentiation and cobblestone morphology, please provide images and immunostaining for the canonical RPE markers BEST1, RPE65, OTX2, MITF, and for comparison use iPSC-RPE.

5. Phagocytosis assay.

The phagocytosis assay used beads which do not accurately predict POS phagocytosis – they should have followed the recommendation of Robin, which stated this correctly.

Phagocytosis assay measurements by flow - Their assay is not adequate to evaluate RPE phagocytosis. They should follow the method in the publication they mis-cited (33).

Issues of data selection: In Figure 4, the authors present flow cytometry data indicating that Ripasudil “significantly enhances phagocytosis in RPE cells, inducing an even greater effect than Y-27632.”

There appear to be no error bars on the figure, how many times was it repeated?

The authors need to show the concentration curves for both drugs, as the improvement of Ripasudil over Y-27632 may just be due to the chosen concentration – as they note. Given that, it is not appropriate to show this graph as it is biased to show findings that fit with their hypothesis.

6. Relevance of ABCA1/4. The authors find a difference in ABCA1 expression. They state “Notably, ABCA1 belongs to the same transporter family as ABCA4, which has been previously identified as a therapeutic target in macular degeneration pathogenesis [39].”

This sentence exhibits some sleight of hand – mutated ABCA4 is a well-known driver of Stargardt’s disease. ABCA4 expressed in photoreceptor cells flips N-retinylidene phosphatidylethanolamine from the luminal to the cytoplasmic side of the membranes of the photoreceptor disc. This is essential for the visual cycle. If ABCA4 is mutated, photoreceptor degeneration can occur. Mutated ABCA4 in the RPE may contribute, but it is the role of ABCA4 in photoreceptors, and particularly in cones, that leads to macular degeneration. This is unlike the mechanism of AMD. It appears the authors are trying to persuade the reader about similarities, which may not be relevant.

ABCA1 stimulates efferocytosis, however in this study it doesn’t do this by promoting phagocytosis, it does this by regulating the release of find-me ligands. PMID: 33741356

Misrepresentation of the literature. "Further connecting these findings to dAMD pathology, Apo-E, a lipid acceptor for ABCA1, has also been identified as a potential therapeutic target for dAMD [40].

I read reference 40 and was unable to find any mention that proposed Apoe as a potential therapeutic target for dry AMD. The authors should provide the quote from the paper that shows their statement is accurate.

10. Given that there is substantial in vitro evidence from multiple publications that ROCK inhibitors can promote RPE phagocytosis, done for example with appropriate RPE and technique compared to their data, in order to move the needle they should test it in the eye in an in vivo model of AMD.

It is important to do this in vivo test because, as mentioned, enhancement of phagocytosis does not entirely make sense.

RPE cells uptake POS and poor digestion of POS leads to build up of debris and drusen. If phagocytosis is enhanced without concomitant increased autophagy, this buildup of undigested POS and drusen could get worse. Hence, they need to show improvements from the treatment in a suitable AMD model in vivo.

11. Reliance on existing literature: The range and potential of Robin's hypotheses are constrained by the existing literature. This naturally limits the ability of Robin to generate truly novel insights. In addition, its reliance on existing literature presents a risk of overfitting analysis to accommodate existing biases and gaps in the current literature. Finch suffers from similar weaknesses in being constrained by established bioinformatic workflows and standardized knowledge and data filtering. This reliance on standardized pipelines may limit its applicability in emerging contexts that are driven by novel analytical strategies used to uncover previously unrecognized biological insights. The authors should clarify the extent to which their current application can address these concerns.

12. Opaque reasoning in pairwise comparison to rank hypotheses:

The inherent complexity of LLMs makes it difficult to interpret or understand the underlying logic used in generating ranked hypotheses from pairwise comparisons. The authors must clarify to what degree end users have flexibility in understanding or adjusting the hypothesis ranking criteria provided by Robin.

In addition, the authors should clarify how their pairwise comparison methodology addresses potential systemic biases, given the limited description of the underlying statistical framework. In particular, prior studies have shown that pairwise comparisons using LLMs may inadvertently favor publications with a more assertive or authoritative tone over those grounded in greater scientific rigor. The authors should explain how the current approach mitigates such biases, or consider alternative comparison methods (i.e. would pointwise comparison make more sense?).

13. Robin's reliance on LLMs introduces sensitivity to prompt phrasing. This renders its candidate ranking system potentially variable and limits the reproducibility of hypothesis prioritization, reducing user confidence in the consistency and robustness of the system's output. For example, the experimental assays recommended by the `robin_full.ipynb` and `robin_demo.ipynb` scripts for dry AMD vary significantly due to changes in search criteria. The authors should provide user guidelines on recommended settings that effectively mitigate the influence of prompt sensitivity on generated results.

14. Bias towards larger effect sizes: The DGE analysis performed by Finch focuses on the identification of a novel risk gene with a 3-fold upregulation. However, it remains unclear whether its standardized bioinformatic pipelines can effectively analyze more complex, multifactorial diseases involving several genes with small individual effect sizes. As an example, demonstrating Finch's accuracy in polygenic risk score analysis could help address this concern.

Minor points

1. Use of the present tense through much of the MS makes it difficult to understand what was done and what could be done.

a. Be clear about what was done by Robin and what was done by experimenters.

2. Although Robin is described as a multi-agent system capable of fully automating key steps of the scientific process, it is better described as a semi-autonomous process.

(Remarks on code availability)

Difficulty in software implementation and virtual environment set-up: While the proposed use of a virtual environment is effective for managing Python-specific dependencies, it may still be susceptible to compatibility issues arising from version mismatches—particularly as newer versions of Python and its associated packages are released. These conflicts can introduce challenges in maintaining reproducibility and long-term stability of the computational environment.

In the course of this review, multiple installation attempts using both PC and Mac systems were attempted, and different bugs/installation issues using one or the other were encountered. This may or may not be due to residual dependencies or interactions between the local host's operating system and the virtual environment. To address this, the authors should consider a more fully self-contained solution such as Docker, which can provide a uniform runtime environment irrespective of the underlying operating system.

Additionally, when setting api keys, a consistent error message was encountered when running the Robin configuration command. It was unable to successfully utilize the `pai` key added to a local `.env` file. Again, this may or may not be due to the local machine setup, but a more self-contained installation program would make installation smoother.

Version 1:

Reviewer comments:

(Remarks to the Author)

The authors have made a substantial effort to address the concerns raised in the first round of review. The addition of primary RPE-SC validation, ablation studies, time-on-task analysis, and agent comparisons represent meaningful improvements. I acknowledge the quality and quantity of new experimental work. However, several significant issues remain that require attention before this manuscript can be recommended for publication.

1. The Finch comparison undermines, rather than supports, the claim that a specialized data analysis agent is necessary.

The manuscript states that Finch demonstrated "equivocal performance" compared to Claude Code and Biomni on both tasks (Section 2.3), and Supplementary Figure S18 visually confirms that Finch and Claude Code achieved near-identical scores on both the RNA-seq and flow cytometry rubrics. The rebuttal reports Finch scores of 85.7% (RNA-seq) and 100% (flow cytometry), but the manuscript does not discuss what "equivocal performance" implies for the system architecture. This is the most important result in the ablation section. If a general-purpose coding agent already matches a purpose-built, containerized bioinformatics agent on the authors' own rubrics — designed by the authors themselves — the justification for Finch as a distinct architectural component is substantially weakened. Furthermore, the comparison is based on only two tasks (RNA-seq differential expression and flow cytometry gating), each with three replicate runs — an evaluation too narrow to draw architectural conclusions. Critically, the flow cytometry task shows a ceiling effect: all three agents achieved 100% on the rubric, meaning the task is too simple to discriminate between a specialized and a general-purpose agent. When benchmark tasks are solved perfectly by all contestants, they cannot inform architectural decisions. The RNA-seq task does not reach a ceiling but presents a separate problem: overlapping standard deviations across agents (Figure S18) make the results statistically indistinguishable. Together, these two issues — one task too easy, the other too noisy to differentiate — mean that neither benchmark supports the claim that Finch offers an advantage. A convincing case for a specialized agent would require evaluation across a broader range of bioinformatics tasks of varying complexity (e.g., single-cell analysis, proteomics, imaging quantification, variant calling) — tasks difficult enough to avoid ceiling effects yet well-defined enough to yield discriminative comparisons. The authors should expand the evaluation to additional task types and directly address what value Finch adds over simply using a general-purpose coding agent within the Robin pipeline.

2. The model comparisons are obsolete for a 2026 submission and must be updated.

The paper uses OpenAI o4-mini as Robin's backbone, Claude Sonnet 3.7 as the LLM judge, and compares against Claude Code (model version unspecified) and OpenAI Deep Research (June 2025). As of the date of this review, the frontier has advanced considerably (Claude Opus 4.5/4.6, GPT-5 class models, Gemini 3.1 Pro). Given that Claude Code already matched Finch with a mid-2025 model (point 1 above), it is reasonable to expect that current-generation general-purpose agents would equal or exceed Finch's performance. Of course, the Robin backbone itself (o4-mini) is now several generations behind — the system's own performance would likely improve substantially with updated models.

For a paper submitted in 2026 to Nature, I consider it essential that the authors update the Finch comparison to include at least one current-generation coding agent (e.g., Claude Code with Opus 4.5 or later). The step-change in coding and analytical capabilities introduced by models such as Claude Opus 4.5 and GPT-5 class systems is well documented and represents a qualitative, not merely incremental, advance over the mid-2025 models used in this study. Given that Claude Code already matched Finch with a weaker model (Figure S18), the outcome with current-generation models bears directly on whether the specialized-agent architecture remains justified. This comparison is straightforward to execute — it requires only providing the same dataset and prompt to an updated agent — and should not be deferred to a Discussion caveat.

3. The biological validation is convincing as a proof-of-concept but should be framed accordingly.

The RPE-SC experiments are a strong addition. The use of primary cells from an aged donor, fluorescently labeled rod outer segments (replacing beads), dose-response curves for both Y-27632 and ripasudil, immunocytochemistry validation of RPE markers, and the identification of KL001 as an additional hit collectively make a compelling *in vitro* case. The ABCA1 upregulation finding and its connection to AMD genetics via ApoE is an interesting mechanistic lead.

However, the authors should be careful not to overstate these findings. This remains a single *in vitro* assay system (phagocytosis in cultured RPE cells), and as Reviewer 2 correctly noted, enhanced phagocytosis without concomitant increases in autophagy could theoretically worsen debris accumulation. The Discussion now appropriately mentions the need for *in vivo* validation, and I appreciate this addition.

4. The time-on-task analysis (Table 1) is a useful addition but contains estimates that should be clearly flagged.

Several entries in Table 1 are derived from published surveys (von Hippel et al. 2015 for hypothesis ideation; Sboner et al. 2011 and the Anaconda Data Science Report for bioinformatics setup) rather than measured directly in the context of this study. While these are reasonable estimates, they should be more prominently acknowledged as approximations to avoid giving the impression that the 200-fold speedup was empirically measured end-to-end.

Overall, the paper presents a genuinely interesting system and the RPE-SC biological validation is solid work. The core architectural idea — connecting literature-grounded hypothesis generation with autonomous experimental data analysis in an iterative loop — is sound and timely. However, the paper's own data (Figure S18) shows no performance advantage of the specialized Finch agent over general-purpose Claude Code, and all model comparisons predate the current generation

of foundation models by a significant margin. These issues bear directly on the claimed contribution of the system architecture and should be addressed in the next revision.

(Remarks on code availability)

My code review findings from the first round — specifically the rate-limiting errors that prevented full reproduction and Finch's restricted availability (not publicly released, accessible only upon request to FutureHouse) — do not appear to have been addressed in this revision.

Referee #2

(Remarks to the Author)

In their responses and revisions, the authors have addressed several key points raised in the initial review, including reframing Robin as a semi-autonomous tool and more clearly acknowledging the role of human input in iterative literature-based discovery. The addition of validation experiments using RPE-SC cells further strengthens the practical utility of the approach. However, several methodological aspects would benefit from additional clarification:

Reviewer pt 9. Opaque reasoning in pairwise comparison to rank hypotheses: The inherent complexity of LLMs makes it difficult to interpret or understand the underlying logic used in generating ranked hypotheses from pairwise comparisons. The authors must clarify to what degree end users have flexibility in understanding or adjusting the hypothesis ranking criteria provided by Robin. In addition, the authors should clarify how their pairwise comparison methodology addresses potential systemic biases, given the limited description of the underlying statistical framework. In particular, prior studies have shown that pairwise comparisons using LLMs may inadvertently favor publications with a more assertive or authoritative tone over those grounded in greater scientific rigor. The authors should explain how the current approach mitigates such biases, or consider alternative comparison methods (i.e. would pointwise comparison make more sense?).

Authors Response to point 9: We thank the reviewer for highlighting the importance of bias mitigation in LLM-based rankings. We are aware of the 'verbosity' and 'tone' biases established in recent literature (e.g., Zheng et al., 2023; Wang et al., 2023), where LLM judges may favor assertive or longer text over factually rigorous content. To mitigate this, our pairwise comparison prompts do not simply ask 'which is better?' but instead utilize a structured rubric that explicitly instructs the model to evaluate specific criteria (e.g., 'Proposed Mechanism Plausibility,' 'Experimental Feasibility') and to penalize vague or purely rhetorical assertiveness. Furthermore, by using our agents (Crow/Falcon) to retrieve the underlying evidence before the ranking step, we ground the comparison in retrieved facts rather than the linguistic style of the hypothesis itself. Importantly, our agents utilize PaperQA2 (Skarlinski et al. 2024), a retrieval augmented generation (RAG) approach which ensures that the claims used to build assay and drug candidate hypotheses are strictly lined to evidence based sources.

Regarding the use of pointwise (single answer) scoring, the same literature (Zheng et al.) shows that pairwise comparisons are generally more robust than pointwise scoring because LLMs lack a stable internal rubric for absolute grading (e.g., a "7/10" means different things in different contexts), whereas they are quite good at relative ranking ("A is better than B"). In general, while pointwise scores are advantageous when criterion-level interpretability or calibrated absolute scores are needed, the literature favors pairwise comparisons for relative ranking of long-form responses under non-adversarial conditions⁴.

Regarding the transparency of the LLM judgments, the LLM judge's reasoning is presented to the user in the "reasoning" column of the CSV files (experimental_assay_ranking_results.csv and therapeutic_candidate_ranking_results.csv).

Transparency is noted as one of the advantages of LLM judges in the Zheng et al. paper:

"LLM-as-a-judge offers two key benefits: scalability and explainability. It reduces the need for human involvement, enabling scalable benchmarks and fast iterations. Additionally, LLM judges provide not only scores but also explanations, making their outputs interpretable..."

Furthermore, the user can read and adjust the rubric used by the LLM judge in prompts.py. The rubrics used in our manuscript are presented in Supplementary Figures S5 and S11.

Reviewer reply to the authors response to point 9:

It remains unclear whether alternative approaches for hypothesis ranking were considered. This is important to clarify as there is no clear consensus in the literature regarding optimal strategies for LLM-based relative rankings. Of particular concern is position bias within pairwise comparison frameworks, and multiple variants exist that incorporate aspects of pointwise comparison to improve robustness (e.g., Choo et al., 2025; Bouchard et al., 2025). Further detail on the authors' choice of methodology would therefore be valuable. Therefore changes should be made to the text.

10. Robin's reliance on LLMs introduces sensitivity to prompt phrasing. This renders its candidate ranking system potentially variable and limits the reproducibility of hypothesis prioritization, reducing user confidence in the consistency and robustness of the system's output. For example, the experimental assays recommended by the robin full.ipynb and robin demo.ipynb scripts for dry AMD vary significantly due to changes in search criteria. The authors should provide user guidelines on recommended settings that effectively mitigate the influence of prompt sensitivity on generated results.

Authors Response to point 10: We agree that stochasticity is inherent to LLM based agents. In the context of scientific discovery, we believe that some degree of stochasticity is beneficial, as it allows the system to explore a broader solution space across multiple runs rather than deterministically converging on the most obvious answers. Additionally, as we argue

in the case of Finch (see point 14 below) it allows the agent to flexibly adapt to a new problem space. We view Robin as a generator of high-quality possibilities that requires expert adjudication, rather than a deterministic, predictable system. It is also worth noting that variability is not unique to AI systems. Human scientific analysis is subject to significant inter- and intra-rater variability; different experts (or the same expert at different times) frequently prioritize different hypotheses when synthesizing vast bodies of literature. We see Robin's variability as a parallel to the diversity of thought found in human collaboration.

We have provided all of the system and user prompts used in this manuscript (Supplementary Figures S2-S11).

Furthermore, these are the default prompts used in robin.full.ipynb and robin.demo.ipynb and do not contain any instructions specific to dry AMD or the phagocytosis assay.

Reviewer Reply to the authors response to point 10:

While stochasticity is an inherent feature of LLMs, the explanation of whether this variability meaningfully parallels human reasoning within the Robin framework remains inadequate. Further prompt-based analyses that distinguishes variability driven by underlying evidence from that arising due to semantic sensitivity would improve user confidence in recommended experiments and hypotheses. Therefore changes should be made to the text.

Remarks on code availability

Reviewers comments: Difficulty in software implementation and virtual environment set-up: While the proposed use of a virtual environment is effective for managing Python-specific dependencies, it may still be susceptible to compatibility issues arising from version mismatches—particularly as newer versions of Python and its associated packages are released. These conflicts can introduce challenges in maintaining reproducibility and long-term stability of the computational environment.

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Additionally, when setting api keys, a consistent error message was encountered when running the Robin configuration command. It was unable to successfully utilize the pai key added to a local .env file. Again, this may or may not be due to the local machine setup, but a more self-contained installation program would make installation smoother.

Reviewer reply to the authors response:

The authors do not appear to have addressed this issue. Our prior comments regarding code reproducibility and implementation variability were not directly addressed, and the associated github repository does not appear to have been updated since the initial review. Clarification on plans for improving reproducibility, such as through containerization of the demo code, would be helpful. Therefore changes should be made to address these comments.

(Remarks on code availability)

please see notes added to the end of the first review. the authors didnt respond to our prior comments so its not clear that they received them.

Version 2:

Reviewer comments:

Referee #1

(Remarks to the Author)

The authors have made a substantial revision within the scope defined by the editorial team. The new BixBench evaluation broadens the Finch assessment beyond the prior two-task comparison and avoids the ceiling effects that weakened the earlier flow-cytometry benchmark. However, the comparison now tests Finch against its base LLM without the same data access or execution environment: Finch receives the capsule data and can execute code in a Jupyter notebook, while Sonnet 3.7 is evaluated as a no-harness text-only baseline. The 22.8% versus 1.6% result therefore supports the general need for a tool-using data-analysis agent, but does not establish that Finch's specialized harness outperforms a general-purpose coding agent operating under matched data-access and execution conditions. I also note in passing that BixBench was developed by the same group (Mitchener et al., FutureHouse — Mitchener is also first author on the present manuscript) and that Finch trajectories execute inside the BixBench-env Docker container, so the benchmark and the system being benchmarked share authorship and infrastructure. The manuscript currently presents BixBench as an external benchmark via reference #29 and does not disclose this overlap. This is not disqualifying — BixBench is publicly released and has been used by external groups, including OpenAI — but the shared provenance should be stated explicitly in the Methods or in a brief note where the benchmark is introduced, so readers can weigh the result accordingly. I note the editorial determination that comparisons against post-submission frontier models and agents are out of scope and I defer to the editors on this point, but the manuscript text should describe the BixBench result more narrowly and avoid implying that it settles the broader architectural question. The added Discussion language acknowledging that general-purpose coding agents may achieve parity with specialized harnesses is appropriate.

Several claim-calibration and consistency issues remain in the manuscript that should be addressed before publication.

First, the abstract continues to overstate what the data support. "First multi-agent system capable of fully automating both hypothesis generation and data analysis," "first AI system to autonomously discover and validate a novel therapeutic candidate," and "promising therapeutic candidate" are all stronger than the proof-of-concept framing the authors adopted elsewhere in the revision. The abstract sentence stating that "all hypotheses, experimental directions, data analyses, and data figures in the main text of this report were produced by Robin" is also inconsistent with the manuscript itself: the Figure 2 and Figure 3 legends explicitly state that the figures were "formatted for readability in publication by a human," the Methods section states that "Read demultiplexing and alignment was performed by a human" prior to Finch's differential expression analysis, and the Results compare Finch's analyses to parallel human analyses (e.g., the 1.89-fold versus 1.75-fold MFI comparison for ripasudil in ARPE-19, and the human flow-cytometry gating shown alongside Finch's for the RPE-SC experiments). This sentence should be removed or narrowed to accurately reflect the distribution of labor between Robin and the human team.

Second, the ABCA1 finding in the RPE-SC validation contains internal inconsistencies that need to be resolved. The Results text and the Supplementary Figure S26 caption both describe ABCA1 as upregulated in ripasudil-treated cells, and on inspection the green ABCA1 marker in the volcano plot does sit at positive log2 fold change, consistent with the upregulation claim. However, the caption reports log2 fold changes of "-0.55 and -0.74" — both negative — which contradicts the positions plotted in the figure and the direction stated in the text. The most likely explanation is a sign error in the reported numerics (the values should presumably read +0.55 and +0.74), but as written the caption is internally contradictory and should be corrected. Separately, even taking the values as positive, |log2FC| of 0.55–0.74 sits below the |log2FC| > 1 significance threshold the manuscript applies elsewhere, including the dashed reference lines visible at ± 1 in Supplementary Figure S26 itself. The qualifier "modestly upregulated" is therefore generous: the effect does not meet the manuscript's own significance criterion, and the very small p-values reflect low within-experiment variance at $n=4$ rather than a biologically meaningful magnitude. This matters because the Results explicitly state that the RPE-SC RNA-seq experiment was performed "to further validate the ABCA1 finding" from the ARPE-19 cells, and a sub-threshold effect of roughly 1.4–1.7 fold is directionally consistent but is not a clean replication of the original 3-fold change ($\log_2\text{FC} \approx 1.6$). The text should be revised to (1) correct the sign error in the caption, and (2) describe the RPE-SC result as directionally consistent with the ARPE-19 finding but substantially smaller in magnitude and below the manuscript's own significance threshold, rather than as an unqualified validation. Reading the manuscript as currently written, the average reader would not realize that the validation experiment produced a sub-threshold effect.

Third, the RPE-SC experiments appear to rely on primary cells from a single aged donor with technical replicates ($n=4$). This is appropriate as an in vitro proof-of-concept, consistent with the revised framing, but the Results and figure captions should explicitly state "one donor, $n=X$ technical replicates" so that readers do not interpret these experiments as donor-level biological validation.

Fourth, the time-on-task analysis contains numbers that are difficult to reconcile at first reading. The Introduction states that Robin analyses "551 papers in 30 minutes compared to an estimated time of 540 hours" for a human; Table 1 reports 551 papers and 359–424 hours for the literature-search line item; and a later Results paragraph reports an end-to-end discovery cycle of "872–937 human hours" and "approximately 825 references." If these represent different workflow scopes (per-run literature search alone versus end-to-end case-study totals, with 825 references being the cumulative count across all rounds rather than the 551-per-run figure), that distinction should be stated explicitly so that the 200-fold speedup claim is not read as internally inconsistent. The 540-hour figure and the 359–424-hour Table 1 figure for what appears to be the same workflow step are particularly hard to reconcile as currently written.

Fifth, the new Methods text on the Bradley–Terry–Luce pairwise comparison framework is a welcome addition, but it asserts that position bias is mitigated "by utilizing a highly distributed tournament paired with the BTL model." BTL aggregates pairwise outcomes and estimates latent strength parameters, but it does not by itself remove systematic positional bias if the LLM judge consistently prefers the first- or second-listed option. The authors should state whether the order of items within each pairwise comparison was randomized or counterbalanced across comparisons; if it was, a single sentence documenting this would resolve the concern.

The remaining points from the prior round have been adequately addressed. The biological framing is appropriately softened: the autophagy claim now requires further validation, ABCA1 is described as having potential implications for dAMD, and references to expediting therapeutic discovery have been removed. The Table 1 caption and accompanying Results text clearly flag the human time-on-task values as estimates derived from published surveys. The open-source release of Finch with a containerized Docker implementation appears to substantially address the earlier reproducibility and installation issues raised in both my first-round review and Reviewer 2's concerns, pending editorial confirmation that the end-to-end loop runs as described. A final copyedit pass is also warranted — the manuscript in its current state contains small errors that are distracting for a Nature submission, including a missing period between the autophagy caveat and the following sentence ("Further work is necessary to validate any effects on autophagy The human version..."), "Figure4E" and "Figure4F" without spaces, doubled punctuation ("(Figure4E)..", "(Figure4F)."), the spelling "phagoytosis" in a figure caption, and "dose dependant" in the Results.

(Remarks on code availability)

Referee #2

(Remarks to the Author)

Our concerns re code availability have been addressed, thank you.

(Remarks on code availability)

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Response to Reviewers for Manuscript “Robin: A Multi-Agent System for Automating Scientific Discovery”

Thank you for taking the time to review our manuscript. We have carefully considered each comment and suggestion and have revised the manuscript accordingly. Below, we detail our responses to each reviewer’s comments point-by-point, referencing the specific edits made in the revised manuscript.

Reviewer 1 Response

1. The claim of being the “first system to fully automate...” is overstated. Several sophisticated multi-agent frameworks have been described recently, including Google’s “AI Co-Scientist,” Boiko et al.’s “Coscientist” (Nature, 2023), and “TxGemma” (Wang et al., 2025, arXiv). Robin’s specific contribution is its focus on biological discovery and its integration of an autonomous data analysis agent (Finch). The claim of primacy should be narrowed to this specific achievement

Response: We have amended our claims to emphasize Robin’s use of an autonomous data analysis agent and its utility in experimental biology which sets it apart from other AI scientists. We have amended the quoted claim to state that Robin is the first multi-agent system to automate “both hypothesis generation and data analysis for experimental biology”. In the introduction we emphasize the unique value of Robin as combining literature-based hypothesis generation and autonomous data analysis in one semi-autonomous system.

Changes: Abstract (Page 1, line 5); Introduction (Page 2, lines 51; 61-64).

2. The novelty of ROCK inhibition is clinically actionable but mechanistically incremental. While repurposing an existing drug is a valuable outcome, the paper would benefit from framing its contribution carefully. The core finding is not the discovery of a new mechanism but rather the identification of a new therapeutic application for a known pathway, given that Y-27632 was previously shown to restore phagocytic efficiency (Mao & Finnemann, 2021).

Response: We agree that ROCK inhibition is a known mechanism of phagocytosis enhancement. While the original manuscript did cite the existing work, we have revised the manuscript to state more clearly that while the mechanism is known, the specific application of ROCK inhibitors for **dry AMD** via the mechanism of **phagocytosis enhancement** is the primary contribution. We explicitly acknowledge previous work on Y-27632 restoring phagocytic efficiency but highlight the gap in applying this to dry AMD pathology.

Changes: Discussion (Page 9, lines 230-236).

3. The reported magnitude of the ripasudil effect requires reconciliation. The manuscript should address the discrepancy between the AI’s reported 7.5-fold increase in MFI and the human analysis showing a 1.75-fold increase, as this points to potential instability in the autonomous analysis pipeline.

Response: The discrepancy was due to Finch reporting the absolute MFI rather than the fold-change. The prompt (made by a human) incorrectly asked it to report the absolute MFI and not the fold-change, so this was a human error and not a Finch error or pipeline instability. To correct this we re-ran Finch with an amended prompt and it correctly reported the fold-change in MFI rather than the absolute MFI. The text now reports a 1.89-fold increase, which aligns with the human analysis (1.75-fold).

Changes: Results Section 2.2.3 (Page 7, Line 174); Figure 4B; Supplementary Figure S10.

4. The paper lacks benchmarking and ablation studies, which are needed to rigorously validate the architecture. The authors should provide: (a) A comparison against a single-agent baseline (e.g., a state-of-the-art model like o3/Gemini 2.5 pro/Claude 4 given the same high-level prompt) (b) Ablation studies that quantify the marginal contribution of each agent (Crow, Falcon, Finch). (c) A head-to-head comparison on an open benchmark eg GPQA to contextualize Robin’s performance against other systems. (d) A runtime and cost analysis (e.g., GPU hours or dollar cost per iteration) to assess scalability.

Response: We have added a new section, “2.3 Ablation studies and validation of the Robin architecture,” to address these points.

- (a) We conducted a head-to-head comparison of Robin with OpenAI’s Deep Research agent. We show that Deep Research failed to identify ROCK inhibition or any other hits in the phagocytosis assay (Supplementary Figure 20).
- (b) We performed ablation studies by replacing Crow and Falcon agents with OpenAI’s o4-mini. We found that ablating these agents led to a dramatic increase in hallucinated references and a significant reduction in proposal quality (Section 2.3 and Supplementary Figure S18).
- (c) LLMs are not able to directly execute code like Finch, so instead of ablating Finch like we did for Crow and Falcon, we conducted head-to-head comparisons of Finch agent against Claude Code and Biomni which demonstrated SOTA performance of Finch on RNA-seq and Flow cytometry data analysis (S19).
- (d) We performed a cost analysis of Robin which is detailed in the results section 2.2.

Robin is specifically designed for iterative biological research and at the time of writing there is no widely accepted benchmark for iterative scientific discovery (GPQA is question-answer style benchmark for general scientific knowledge and reasoning). We believe this real-world assessment and the above ablation and comparison studies provide a more relevant assessment of the system’s validity. For benchmarking of the PaperQA2 language agent on which Crow and Falcon are built, see Skarlinski et al. 2025.

Changes: Results Section 2.3; Methods Section 3.3; Supplementary Figures S18, S19 and S20.

5. The biological validation would benefit from more translationally relevant models. The work relies solely on immortalized ARPE-19 cells. Because dry-AMD pathogenesis is still poorly understood, the proposed mechanism may ultimately fall short in a clinical setting. Including validation in primary RPE cells, retinal organoids, or an in-vivo model would strengthen the findings.

Response: We have included new experimental validation using **primary Retinal Pigment Epithelium Stem Cells (RPE-SC)** derived from an aged donor and expanded for no more than 2 passages *in vitro*. We validated the RPE phenotype of these cells by immunocytochemistry, functional VEGF polarization and morphology (Supplementary Figures S24 and S25).

All of the drugs screened in ARPE-19 cells were again screened in RPE-SC in addition to drugs chosen by OpenAI's Deep Research for comparison. The results in RPE-SC confirmed the hits from the ARPE-19 screen. Furthermore, Ripasudil showing greater potency than Y-27632 in a dose response experiment conducted in both ARPE-19 and RPE-SC. Of note, the RPE-SC drug screen found an additional hit not seen in the ARPE-19 screen (KL001, a cryptochrome stabilizer suggested by Robin. To our knowledge, this is a completely novel finding).

Changes: Results Section 2.2.4 (Page 9, Lines 178-200); Figure 4D-E; Supplementary Figures S20, S24, S25, S26.

6. The generalization claims are illustrative rather than validated. The paper lists 10 further diseases with candidate tables, but provides no experimental follow-up, making them unproven examples of the system's ideation capability.

Response: Agreed. We have removed the supplementary figures which contain illustrative candidate drugs for ten diseases.

Changes: Supplementary material.

7. The statistical reporting should be more detailed. To clarify the experimental findings, the authors should report the number of biological and technical replicates for each assay, include raw effect sizes, and provide adjusted P-values.

Response: We have updated the figure captions to include the number of technical replicates, p-values and statistical tests performed and been more clear about statistical methods in the methods section. We have added the raw MFI measurements and the plotted fold-changes for the ARPE-19 and RPE-SC screening experiments as supplementary tables. Additionally, the dose response experiments in Figure 4C and 4E display the MFI. Note that the raw mean fluorescence intensity (MFI) values from flow cytometry data are a function of how the user (or agent) plots the gates and are not directly comparable between experiments.

Changes: Figures 2, 3, 4 Captions; Supplementary Figure S21 and S22 Captions.

8. The reproducibility of the Finch agent is a concern, and its stochastic nature would benefit from further characterization with run-to-run variance metrics.

Response: We show in Figure 3C that Finch identified the same differentially expressed genes in over 50% of trajectories. Additionally, we performed a comparison of Finch's performance with a leading AI science agent and a coding agent (Biomni and Claude

Code, respectively). In the comparison with Biomni and Claude Code, (Supplementary Figure 19), Finch consistently achieved scores of 85.7% and 100% on the rubrics for RNA-seq analysis and flow cytometry analysis, respectively (see Supplementary Figures 12, 13 and 19).

Changes: Results Section 2.3 (Page 9, Lines 216-219); Methods Section 3.2.2 (page 11, lines 313 - 316); Supplementary Figures 12, 13 and 19.

9. The human effort involved is not quantified. To substantiate claims of automation, the authors should provide a time-on-task table detailing the minutes of scientist involvement required for each stage of the discovery loop.

Response: We appreciate the reviewer’s suggestion to quantify the human involvement required by the Robin framework. We have added a “time-on-task” analysis (Table 1) to substantiate our claims of automation. This analysis estimates the manual effort required to replicate Robin’s literature synthesis and data analysis steps. Notably, in a typical discovery run, Robin’s agents (Crow and Falcon) synthesize an average of 551 unique references—a volume of scholarly work that would require a human researcher approximately 540 hours of manual reading and synthesis. By automating these cognitive bottlenecks, Robin could reduce the total “cognitive time” of the discovery loop from hundreds of hours to less than two hours of high-level supervision. We have updated the manuscript to consistently describe Robin as a “semi-autonomous” and “lab-in-the-loop” system, emphasizing that human effort is redirected from manual data processing toward experimental validation.

Changes: Results Section 2.1 (Lines 79-81); Table 1.

10. The data hand-off between agents is opaque. The manuscript does not provide the precise, machine-readable artifact that passes Finch’s analysis to Crow/Falcon. Providing a concrete example of that prompt or data structure is necessary to assess the degree of automation versus hidden human curation.

Response: We have updated Figure 4A to include a more detailed workflow illustrating the hand-off between Finch and Crow and added additional explanation in the Methods section. In summary, Finch hands-off the results of its analysis to an LLM (o4-mini) as a CSV file along with a prompt. The prompt is a JSON template which tells the LLM to create a structured output from the Finch results including what experiments have been conducted, a summary of the results, insights from the results and important questions that have been raised. This prompt can be found in prompts.py in the Robin GitHub repository as the variable ‘EXPERIMENTAL_INSIGHTS_APPENDAGE’. This structured output is used by Robin to generate new candidate ideas and query generation prompts (for background literature search).

Changes: Figure 4A and Caption.

11. A literature quality-control filter is needed. To mitigate the “garbage-in, garbage-out” risk, the system would be improved by having the Crow agent assign a confidence score to sources (e.g., based on citation count or journal impact) before they are consumed by other agents.

Response: The agents Crow and Falcon are based on the PaperQA2 literature agent developed at FutureHouse (Skarlinski et al. 2024). Metadata (citation count and journal

title) is used by PaperQA2 during the Reranking and Contextual Summarization (RCS) step. This metadata is provided to the underlying LLM to help it weigh the authority and impact of evidence when formulating a response. Additional measures (citation traversal and relevance scoring) are used to prioritize sources by relevance (Skarlinski et al. 2024). Additionally, in our new ablation study (Section 2.3), we demonstrate that the Crow and Falcon literature search agents produce almost no "hallucinated" references compared to a frontier LLM (o4-mini).

12. The manuscript should ideally include a discussion of ethics and safety, noting that automated discovery pipelines should incorporate guardrails to address potential dual-use concerns or off-target effects.

Response: We agree that safety guardrails are essential for automated discovery systems. We interpret "dual-use" as the risk of a system being repurposed to design harmful biological agents, and "off-target effects" as the risk of proposing toxic therapeutic candidates.

To address off-target risks, we explicitly incorporated safety constraints into Robin's ranking criteria (Supplementary Figure S11); the system is instructed to prioritize candidates with known safety profiles and flag potential toxicity as a negative criterion. Furthermore, as a "lab-in-the-loop" framework, Robin generates therapeutic candidates that are subject to traditional pre-clinical safety filters. Any toxicity or off-target effects would be identified through the standard progression of cell assays and animal experiments that are required before any human exposure occurs.

Regarding dual-use concerns, Robin utilizes foundation models (e.g., GPT-4o, Gemini) that have undergone extensive safety alignment (RLHF) to prevent the generation of malicious biological protocols. Additionally, all queries to the FutureHouse/Edison Scientific platform are run through an LLM classifier that checks against unsafe topics. Repeat violaters are banned from using the platform. Furthermore, the system's current architecture is focused on synthesizing existing literature and specific datasets, which limits its utility for the *de novo* design of hazardous materials.

We have added a paragraph in the discussion section to address concerns regarding dual-use and off-target effects.

Changes: Discussion (Page 9, Lines 247-254)

Reviewer 2 Response

1. Novelty claim: Abstract:

“Statement that we were able to identify a novel treatment for dry age-related macular degeneration (dAMD), the major cause of blindness in the developed world. This is far overstated. They have not discovered a treatment for dry AMD and should be edited. Similarly, the statement Ripasudil is a clinically-used rho kinase (ROCK) inhibitor that has never previously been proposed for treating dAMD.”

While this may be the case for this specific ROCK inhibitor, the concept that ROCK inhibition might benefit dry AMD is in the literature, as examples...

Response: Thank you for your comments, which have improved the quality of our manuscript. We have revised the abstract to state that we identified a “promising therapeutic candidate” rather than a “treatment.” Additionally, we have updated the Discussion to explicitly acknowledge that while ROCK inhibitors have been suggested for wet AMD and other retinal diseases, Robin’s contribution is proposing them for dry AMD specifically via the mechanism of **phagocytosis enhancement**. We have cited relevant literature to support the distinction.

While others have proposed ROCK inhibition for several ocular disease including wet and dry macular degeneration, no one to date (including the example studies you presented) has proposed ROCK inhibition as a treatment for dry AMD via a mechanism of phagocytosis enhancement. From the examples you kindly shared the proposed mechanism are:

- PMID: 40250716 – ROCK inhibition for dAMD via anti-inflammatory effects,
- PMID: 27917311 – promotion of wound healing,
- PMID: 39594655 – autophagic vacuole formation,
- PMID: 32252692 – inflammation via the miR-136-5p/ROCK/NLRP3 pathway.

The distinct proposed mechanism of action is essential as it provides a route for laboratory validation, informs safety and toxicity of the drug, informs clinical trial design and improved the patentability of the idea. As we expand on below (see point 11) and in the manuscript (see Discussion, lines 230-241), LLM agents are ideal for recombining established ideas into novel and testable hypotheses.

Changes: Abstract (Page 1, Line 8); Discussion (Page 9, Lines 230 - 232)

2. Novelty claim: “Notably, while ROCK inhibitors have been previously suggested for treatment of wet AMD and other retinal diseases of neovascularization, Robin is the first to propose their application in dry AMD for their effect on phagocytosis [35]. This approach is supported by several lines of evidence, as RPE phagocytic dysfunction is observed not only with normal aging but is pronounced in AMD patients [41, 42, 43].”

35) ROCK inhibitors in ophthalmology: A critical review of the existing clinical evidence

42) PMID: 37371481. This paper considers stimulating RPE phagocytosis and autophagy in dry AMD treatments and explains why targeting phagocytosis is likely not the best route. This demonstrates that increasing phagocytosis as an AMD treatment has been considered in the literature.

PMID: 33024620. Ripadusil increased phagocytosis in trabecular meshwork cells in pig eyes.

Authors state: “Preclinical models have demonstrated that Y-27632 can restore phagocytic efficiency in RPE cells [33], confirming Robin’s literature-based rationale for suggesting this candidate.”

I checked reference 33, there is no mention of Y-27632.

Response: We have considered the three papers shared:

PMID: 37037790 This review does not discuss treating dAMD by enhancing RPE phagocytosis via ROCK inhibition.

PMID: 37371481 While this review article considers phagocytosis enhancement in AMD, it does not identify rho-kinase as a target or mention ROCK inhibition as a potential mechanism of action. Instead, CD36 is identified as a target for phagocytosis enhancement. This review claims that phagocytosis has been previously considered as a target in AMD but provides no supporting literature to evidence this claim. The claims this review article makes about autophagy being a preferable target over phagocytosis are not either supported or directly contradicted by the literature cited in the paper. For example:

“First, the autophagy pathway can avoid injuring healthy cells through its self-regulation mechanisms, thereby reducing the risks of adverse side effects and toxicity. Second, the clearance efficiency of the autophagy pathway is higher than that of phagocytosis, allowing for rapid removal of waste materials and maintenance of cellular homeostasis and normal metabolic function. Third, phagocytosis can lead to the production of many inflammatory cells, while the autophagy process is relatively stable and does not cause the release of large amounts of cytokines, thereby avoiding the occurrence of inflammatory reactions [151,152].”

None of the three claims this review (PMID: 37371481) makes about the relative merits of autophagy over phagocytosis as a drug target are supported by the cited references (references 151 and 152). In fact, the third claim is directly contradicted by reference 151 which explains that secretory autophagy is involved in the secretion of leaderless cytosolic proteins, including pro-inflammatory cytokines like IL1B and IL18. Furthermore, the wider literature does not support these claims. For example, there is no evidence that phagocytic bursts, which occur diurnally in human RPE, lead to the recruitment of many inflammatory cells to the retina.

Looking at each claim individually:

Claim 1. The literature does not demonstrate that autophagy generally “avoids harming healthy cells” or reduces systemic toxicity across contexts; effects are context dependent and autophagy can, in certain settings (e.g., secretory autophagy), facilitate release of inflammatory mediators ¹.

¹Deretic, V. Autophagy in inflammation, infection, and immunometabolism. Immunity 54, 437–453 (2021).

Claim 2. Autophagy and phagocytosis have different scopes and substrates: autophagy degrades intracellular cargo via autophagosome–lysosome fusion, whereas phagocytosis/efferocytosis removes extracellular targets (apoptotic corpses, microbes) and, in professional phagocytes, operates with very high engulfment capacity. The literature provides no head-to-head quantitative evidence that autophagy is faster or more efficient than phagocytosis ².

Claim 3. This is an oversimplified and incorrect blanket statement. Autophagy often dampens inflammation by removing DAMPs and directly targeting inflammasome components. Yet autophagy can also participate in secretory pathways that export IL-1 β /HMGB1. Phagocytosis can be anti-inflammatory in the context of efferocytosis, thus promoting resolution, but can be pro-inflammatory when PRR signaling recognizes pathogens. LC3-associated processes (e.g., LAP/LANDO) bridge autophagy and phagocytosis and can be anti-inflammatory. Thus, effects on inflammation depend on cargo, receptors, and cell type; a categorical dichotomy is not supported ³.

PMID: 33024620 This study investigates the utility of Ripasudil in pigmentary glaucoma, not age related macular degeneration. The cell types studied (trabecular meshwork cells) while being in the eye, are not present in the retina and are not known to be directly involved in macular degeneration.

Reference [33] Regarding reference [33] in the original Robin manuscript: we mistakenly cited Mao and Finneman 2012 instead of Mao and Finnemann 2021. This error has been corrected.

3. Novelty claim: ‘In investigating the basis for Robin’s initial identification of Y-27632, we found that the literature search included a single paper demonstrating Y-27632’s ability to enhance phagocytic efficiency in ‘low-phagocytic’ RPE cells from human donors by promoting actin polymerization [42].’ 42 – I looked at that citation and could not find any reference to Y-27632. Hence, again, this is a mis-cited reference. However this is described in PMID: 34440696, and ROCK inhibition stimulating RPE phagocytosis is also described in PMID: 29456184 and PMID: 40389756. Hence, there are at least 3 relevant papers that the authors should have cited. Therefore the authors claim the novelty of their findings by not fully describing the literature.

Response: Reference [42] was mistakenly cited in the original manuscript and should have cited Mao and Finnemann 2021, as with the previous reference error. There was no attempt to misrepresent the novelty of our findings. For example, in section 2.2.2 of the original manuscript we reference Mao and Finnemann 2021 with regards to Y-27632’s effect on phagocytosis:

”While previous studies have shown that Y-27632 enhances phagocytic cup formation through post-translational regulation of F-actin dynamics [34, 35, 36, 37, 38...”

²Khandia, R. *et al.* A Comprehensive Review of Autophagy and Its Various Roles in Infectious, Non-Infectious, and Lifestyle Diseases: Current Knowledge and Prospects for Disease Prevention, Novel Drug Design, and Therapy. *Cells* 8, (2019).

³Deretic, V. Autophagy in inflammation, infection, and immunometabolism. *Immunity* 54, 437–453 (2021).

We have corrected the citation error in the original manuscript by replacing the original reference [42] with references which demonstrate Y-27632's effect on phagocytosis.

Changes: Discussion (Page 9, Lines 233-234).

4. Validation using RPE cells. The authors should also have followed Robin's direction to use PSC-RPE as well as ARPE19 cells. Instead, they just used ARPE19 cells, which are not a good model of authentic RPE, having a different transcriptome, morphology, and phagocytosis. Moreover, there is considerable variation in the line, depending on the source and how it is grown. This paper provides guidance on what should be provided in publications using ARPE19 cells: PMID: 35314139. They should submit the information requested in that publication. For example, they claim that the cells show signs of differentiation and cobblestone morphology, please provide images and immunostaining for the canonical RPE markers BEST1, RPE65, OTX2, MITF, and for comparison use iPSC-RPE.

Response: We have addressed this by performing validation experiments in **primary human RPE-SC** cells, derived from an aged donor. We also provide immunocytochemistry data confirming the expression of canonical RPE markers (BEST1, RPE65, OTX2, MITF), phagocytic signaling proteins (Integrin $\alpha_v\beta_5$ and the MER proto-oncogene) and morphological characterization in the supplementary material to validate the model. As expected, the morphology of RPE-SC cells is closer to the native form than the ARPE-19 cell line.

Changes: Results Section 2.2.4 (Page 9); Figure 4D-E (Page 7); Supplementary Figures S23 and S24 (Pages 39, 40).

5. The phagocytosis assay used beads which do not accurately predict POS phagocytosis – they should have followed the recommendation of Robin, which stated this correctly. Phagocytosis assay measurements by flow - Their assay is not adequate to evaluate RPE phagocytosis. They should follow the method in the publication they mis-cited (33).

Issues of data selection: In Figure 4, the authors present flow cytometry data indicating that ripasudil “significantly enhances phagocytosis in RPE cells, inducing an even greater effect than Y-27632.” There appear to be no error bars on the figure, how many times was it repeated? The authors need to show the concentration curves for both drugs, as the improvement of ripasudil over Y-27832 may just be due to the chosen concentration – as they note. Given that, it is not appropriate to show this graph as it is biased to show findings that fit with their hypothesis.

Response: We agree with the reviewer regarding the limitations of beads. In the validation experiments with RPE-SC (Section 2.2.4), we replaced pHrodo beads with **fluorescently labeled bovine rod outer segments (ROS)**, as suggested by Robin and the reviewer, to provide a more physiologically relevant assay. We repeated the drug screen (all drugs screened in round 1 and round 2) in RPE-SC cells with ROS and found that ripasudil has the highest effect on phagocytosis of the drugs screened. This effect was dose-dependent in both ARPE-19 cells and RPE-SC.

Changes: Results Section 2.2.4 (Page 9); Methods Section 3.5.3 (Page 13).

6. Relevance of ABCA1/4. The authors find a difference in ABCA1 expression. They state “Notably, ABCA1 belongs to the same transporter family as ABCA4,

which has been previously identified as a therapeutic target in macular degeneration pathogenesis [39].” This sentence exhibits some sleight of hand – mutated ABCA4 is a well-known driver of Stargardts disease. ABCA4 expressed in photoreceptor cells flips N-retinylidene phosphatidylethanolamine from the luminal to the cytoplasmic side of the membranes of the photoreceptor disc. This is essential for the visual cycle. If ABCA4 is mutated, photoreceptor degeneration can occur. Mutated ABCA4 in the RPE may contribute, but it is the role of ABCA4 in photoreceptors, and particularly in cones, that leads to macular degeneration. This is unlike the mechanism of AMD. It appears the authors are trying to persuade the reader about similarities, which may not be relevant. ABCA1 stimulates efferocytosis, however in this study it doesn’t do this by promoting phagocytosis, it does this by regulating the release of find-me ligands. PMID: 33741356

Misrepresentation of the literature. “Further connecting these findings to dAMD pathology, Apo-E, a lipid acceptor for ABCA1, has also been identified as a potential therapeutic target for dAMD [40]. I read reference 40 and was unable to find any mention that proposed Apoe as a potential therapeutic target for dry AMD. The authors should provide the quote from the paper that shows their statement is accurate.

Response: We have corrected the text regarding Apo-E and ABCA1. The revised manuscript states that Apo-E has been identified as a “genetic susceptibility allele” rather than a “therapeutic target,” citing references [42, 43, 44]. This removes the implication that we are claiming it as a direct drug target for dAMD in this context.

Changes: Results Section 2.2.2 (Page 7, Lines 167-169).

7. Given that there is substantial in vitro evidence from multiple publications that ROCK inhibitors can promote RPE phagocytosis, done for example with appropriate RPE and technique compared to their data, in order to move the needle they should test it in the eye in an in vivo model of AMD. It is important to do this in vivo test because, as mentioned, enhancement of phagocytosis does not entirely make sense. RPE cells uptake POS and poor digestion of POS leads to build up of debris and drusen. If phagocytosis is enhanced without concomitant increased autophagy, this buildup of undigested POS and drusen could get worse. Hence, they need to show improvements from the treatment in a suitable AMD model in vivo.

Response: We acknowledge the necessity of in vivo validation. We have added text to the Results and Discussion stating that these hypotheses “would of course require validation in a suitable disease model and ultimately in a randomized, placebo-controlled trial.”

Changes: Results Section 2.2.4 (Page 9, Lines 205-206); Discussion (Page 10, Lines 246-248).

8. Reliance on existing literature: The range and potential of Robin’s hypotheses are constrained by the existing literature. This naturally limits the ability of Robin to generate truly novel insights. In addition, its reliance on existing literature presents a risk of overfitting analysis to accommodate existing biases and gaps in the current literature. Finch suffers from similar weaknesses in being constrained by established bioinformatic workflows and standardized knowledge and data filtering. This reliance on standardized pipelines may limit its applicability in emerging contexts that are driven by novel analytical strategies used to uncover previously unrecognized biological

insights. The authors should clarify the extent to which their current application can address these concerns.

Response: We appreciate the reviewer’s comment regarding the boundaries of AI-driven discovery. We agree that Robin and Finch operate primarily within the bounds of existing knowledge and established methodologies. However, we argue that this constraint is also the system’s primary strength in the context of finding **low-hanging fruit drug repurposing** opportunities:

While Robin only primarily knowledge from the existing literature, it excels at combinatorial synthesis; identifying non-obvious connections between disparate fields that human experts may overlook due to the sheer volume of publications and compartmentalization of knowledge. The proposal of ripasudil for dAMD is a prime example: the drug existed and its effect on phagocytosis was known, but it had not been proposed as a treatment for dAMD by a mechanism of phagocytosis enhancement. We posit that a significant portion of “low-hanging fruit” in therapeutics discovery lies in these unmapped intersections of existing knowledge (i.e., drug repurposing), which LLM-based agents are uniquely positioned to uncover. We give more examples of novelty by synthesis in the introduction.

Regarding Finch, the reliance on standardized bioinformatic workflows (e.g., standard differential expression pipelines) is an advantage as it prioritizes reproducibility and robustness. In an autonomous setting, the risk of “p-hacking” or analytical hallucination is higher if the agent is given complete freedom to invent novel statistical approaches. Additionally, novel statistical approaches require validation. For a system designed to find low-hanging fruit drug opportunities, it is not advantageous to spend time validating novel statistical approaches.

Similarly, Crow is designed to retrieve and propose established, off-the-shelf assays (e.g., the pHrodo phagocytosis assay) rather than novel experimental protocols. This ensures that the “lab-in-the-loop” resources are focused on validating the therapeutic hypothesis, rather than spending time validating an unproven assay.

Finally, by grounding the candidate search in existing literature, the system naturally prioritizes compounds with established safety and efficacy data (drug repurposing). This constraint effectively de-risks the discovery process: it favors candidates like ripasudil that have a clearer path to clinical application, avoiding the high attrition rates associated with characterizing completely novel chemical entities. One could imagine this utility extending to fields such as materials science (generating polymers that have a desired set of properties) or agrochemicals (identifying existing, safe compounds that have antifungal or pesticidal properties).

We have updated the discussion to more clearly explain the merits and constraints of Robin, and to frame the Ripasudil hypothesis in the context of the existing literature.

Changes: Discussion (Page 9, Lines 230-241).

9. Opaque reasoning in pairwise comparison to rank hypotheses: The inherent complexity of LLMs makes it difficult to interpret or understand the underlying logic used in generating ranked hypotheses from pairwise comparisons. The authors must clarify to what degree end users have flexibility in understanding or adjusting the hypothesis ranking criteria provided by Robin. In addition, the authors should clarify how their

pairwise comparison methodology addresses potential systemic biases, given the limited description of the underlying statistical framework. In particular, prior studies have shown that pairwise comparisons using LLMs may inadvertently favor publications with a more assertive or authoritative tone over those grounded in greater scientific rigor. The authors should explain how the current approach mitigates such biases, or consider alternative comparison methods (i.e. would pointwise comparison make more sense?).

Response: We thank the reviewer for highlighting the importance of bias mitigation in LLM-based rankings. We are aware of the 'verbosity' and 'tone' biases established in recent literature (e.g., **Zheng et al., 2023**; **Wang et al., 2023**), where LLM judges may favor assertive or longer text over factually rigorous content.

To mitigate this, our pairwise comparison prompts do not simply ask 'which is better?' but instead utilize a structured rubric that explicitly instructs the model to evaluate specific criteria (e.g., 'Proposed Mechanism Plausibility,' 'Experimental Feasibility') and to penalize vague or purely rhetorical assertiveness. Furthermore, by using our agents (Crow/Falcon) to retrieve the underlying evidence *before* the ranking step, we ground the comparison in retrieved facts rather than the linguistic style of the hypothesis itself. Importantly, our agents utilize PaperQA2 (Skarlinski et al. 2024), a retrieval augmented generation (RAG) approach which ensures that the claims used to build assay and drug candidate hypotheses are strictly lined to evidence based sources.

Regarding the use of pointwise (single answer) scoring, the same literature (Zheng et al.) shows that pairwise comparisons are generally more robust than pointwise scoring because LLMs lack a stable internal rubric for absolute grading (e.g., a "7/10" means different things in different contexts), whereas they are quite good at relative ranking ("A is better than B"). In general, while pointwise scores are advantageous when criterion-level interpretability or calibrated absolute scores are needed, the literature favors pairwise comparisons for relative ranking of long-form responses under non-adversarial conditions⁴.

Regarding the transparency of the LLM judgments, the LLM judge's reasoning is presented to the user in the "reasoning" column of the CSV files (**experimental_assay_ranking_results.csv** and **therapeutic_candidate_ranking_results.csv**). Transparency is noted as one of the advantages of LLM judges in the Zheng et al. paper:

"LLM-as-a-judge offers two key benefits: scalability and explainability. It reduces the need for human involvement, enabling scalable benchmarks and fast iterations. Additionally, LLM judges provide not only scores but also explanations, making their outputs interpretable..."

Furthermore, the user can read and adjust the rubric used by the LLM judge in **prompts.py**. The rubrics used in our manuscript are presented in Supplementary Figures S5 and S11.

⁴Liu, Y. *et al.* Aligning with Human Judgement: The Role of Pairwise Preference in Large Language Model Evaluators. Preprint at <https://doi.org/10.48550/arXiv.2403.16950> (2025); Wang et al. Improving LLM-as-a-Judge Inference with the Judgment Distribution. Preprint at <https://doi.org/10.48550/arXiv.2503.03064> (2025); Liusie, A et al. LLM Comparative Assessment: Zero-shot NLG Evaluation through Pairwise Comparisons using Large Language Models. in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)* (eds Graham, Y. & Purver, M.) 139–151 (Association for Computational Linguistics, St. Julian's, Malta, 2024). doi:10.18653/v1/2024.eacl-long.8.

10. Robin’s reliance on LLMs introduces sensitivity to prompt phrasing. This renders its candidate ranking system potentially variable and limits the reproducibility of hypothesis prioritization, reducing user confidence in the consistency and robustness of the system’s output. For example, the experimental assays recommended by the `robin_full.ipynb` and `robin_demo.ipynb` scripts for dry AMD vary significantly due to changes in search criteria. The authors should provide user guidelines on recommended settings that effectively mitigate the influence of prompt sensitivity on generated results.

Response: We agree that stochasticity is inherent to LLM based agents. In the context of scientific discovery, we believe that some degree of stochasticity is beneficial, as it allows the system to explore a broader solution space across multiple runs rather than deterministically converging on the most obvious answers. Additionally, as we argue in the case of Finch (see point 14 below) it allows the agent to flexibly adapt to a new problem space. We view Robin as a generator of high-quality possibilities that requires expert adjudication, rather than a deterministic, predictable system. It is also worth noting that variability is not unique to AI systems. Human scientific analysis is subject to significant inter- and intra-rater variability; different experts (or the same expert at different times) frequently prioritize different hypotheses when synthesizing vast bodies of literature. We see Robin’s variability as a parallel to the diversity of thought found in human collaboration.

We have provided all of the system and user prompts used in this manuscript (Supplementary Figures S2-S11). Furthermore, these are the default prompts used in `robin_full.ipynb` and `robin_demo.ipynb` and do not contain any instructions specific to dry AMD or the phagocytosis assay.

11. Bias towards larger effect sizes: The DGE analysis performed by Finch focuses on the identification of a novel risk gene with a 3-fold upregulation. However, it remains unclear whether its standardized bioinformatic pipelines can effectively analyze more complex, multifactorial diseases involving several genes with small individual effect sizes. As an example, demonstrating Finch’s accuracy in polygenic risk score analysis could help address this concern.

Response: We appreciate the reviewer’s concern regarding the potential bias towards large effect sizes in our demonstration. We would like to clarify that the ABCA1 upregulation discussed in our manuscript was an emergent result of that specific dataset and the agent’s initial findings, rather than a hard-coded constraint of the system.

We can address this concern by highlighting the fundamental difference between Finch and standard “hard-coded” bioinformatic pipelines:

1. **Finch is a coding agent, not a static pipeline:** Unlike traditional automated pipelines that rely on pre-set thresholds (e.g., $\log_2\text{FC} > 1.5$), Finch operates as a language-model-based agent that **generates and executes arbitrary Python/R code** in a Jupyter environment⁵. It is not restricted to a single workflow. If a user asks for a Polygenic Risk Score (PRS) analysis or if the data

⁵Mitchener et al., “BixBench: a Comprehensive Benchmark for LLM-based Agents in Computational Biology,” *arXiv preprint arXiv:2503.00096* (2025).

suggests weak individual effect sizes, Finch can draft and execute the appropriate code to handle those statistical requirements (e.g., using `scikit-learn` for complex modeling or specific GWAS packages).

2. Iterative reasoning & statistical flexibility: Finch’s architecture is designed for **iterative reasoning**. It observes the output of its code and refines its approach. *Scenario:* In a multifactorial disease context where a simple Differential Gene Expression (DGE) analysis yields no significant hits due to small effect sizes, Finch can be prompted (or self-prompt) to pivot to pathway enrichment analysis (GSEA) or PRS calculation to capture aggregate effects, just as a human bioinformatician would.

Evidence of this capability is already documented in the recent deployment of Finch within the Kosmos framework⁶. Specifically, in Discovery 4, the agent successfully utilized Mendelian Randomization on GWAS summary statistics—an analysis that inherently relies on aggregating small genetic effect sizes rather than large fold-changes—to identify SOD2 as a causal driver of myocardial fibrosis. Similarly, in Discovery 6, the agent autonomously constructed a segmented regression model to detect subtle temporal breakpoints in Alzheimer’s disease progression, confirming its ability to move beyond standard differential expression pipelines. Direct hyperlinks the Finch analysis trajectories are available in Tables 1 and 2 of the Kosmos preprint.

We have clarified the nature of Finch more clearly in the Methods section and added a reference to the Aviary framework⁷ preprint which Finch utilizes.

Changes: Methods Section 3.2 (Page 10, Lines 291-293).

Minor points

Response:

1. We have rectified the use of the present tense throughout the manuscript to clarify what Robin actually did in this manuscript.
2. We have clarified the distinction between Robin’s actions and human actions throughout the text and figure captions.
3. We have updated the manuscript to use the term “semi-autonomous” to accurately reflect the human-in-the-loop nature of the system.

Changes: Abstract (Page 1, Line 53); Introduction (Page 2, Lines 51-62); Results (lines 72-129); all figure captions.

⁶Mitchener et al., “Kosmos: An AI Scientist for Scientific Discovery,” arXiv preprint arxiv.org/pdf/2511.02824 2025.

⁷Narayanan et al., “AVIARY: Training Language Agents on Challenging Scientific Tasks,” *arXiv preprint arXiv:2412.21154* (2024).

Second Response to Reviewers for Manuscript “Robin: A Multi-Agent System for Automating Scientific Discovery”

Thank you for taking the time to review our revised manuscript. Below, we detail our responses to the remaining concerns point-by-point, referencing the specific edits made in the newly revised manuscript.

Reviewer 1 Response

1. The Finch comparison undermines, rather than supports, the claim that a specialized data analysis agent is necessary. The manuscript states that Finch demonstrated “equivocal performance” compared to Claude Code and Biomni on both tasks (Section 2.3), and Supplementary Figure S18 visually confirms that Finch and Claude Code achieved near-identical scores on both the RNA-seq and flow cytometry rubrics. The rebuttal reports Finch scores of 85.7% (RNA-seq) and 100% (flow cytometry), but the manuscript does not discuss what “equivocal performance” implies for the system architecture. This is the most important result in the ablation section. If a general-purpose coding agent already matches a purpose-built, containerized bioinformatics agent on the authors’ own rubrics — designed by the authors themselves — the justification for Finch as a distinct architectural component is substantially weakened. Furthermore, the comparison is based on only two tasks (RNA-seq differential expression and flow cytometry gating), each with three replicate runs — an evaluation too narrow to draw architectural conclusions. Critically, the flow cytometry task shows a ceiling effect: all three agents achieved 100% on the rubric, meaning the task is too simple to discriminate between a specialized and a general-purpose agent. When benchmark tasks are solved perfectly by all contestants, they cannot inform architectural decisions. The RNA-seq task does not reach a ceiling but presents a separate problem: overlapping standard deviations across agents (Figure S18) make the results statistically indistinguishable. Together, these two issues — one task too easy, the other too noisy to differentiate — mean that neither benchmark supports the claim that Finch offers an advantage. A convincing case for a specialized agent would require evaluation across a broader range of bioinformatics tasks of varying complexity (e.g., single-cell analysis, proteomics, imaging quantification, variant calling) — tasks difficult enough to avoid ceiling effects yet well-defined enough to yield discriminative comparisons. The authors should expand the evaluation to additional task types and directly address what value Finch adds over simply using a general-purpose coding agent within the Robin pipeline.

Response: We appreciate the reviewer’s critical evaluation of the Finch comparisons, and we agree that measuring Finch’s performance only on the tasks required for the

biological work in the manuscript does not accurately characterize the overall capabilities of Finch. To address this, we updated the manuscript to include Finch’s performance on 170 tasks related to human biology and drug discovery from the BixBench bioinformatics benchmark, which is widely used to assess the performance of LLMs on bioinformatics tasks (see this OpenAI press release for a recent example of its use: <https://openai.com/index/introducing-gpt-rosalind/>). Of note, this benchmark was also available at the time of our initial manuscript submission. We compared Finch’s performance to Claude Sonnet 3.7, the LLM used in Finch at the time this manuscript was submitted. This comparison demonstrates the utility of the Finch agentic coding harness, as Finch achieves an overall score of 23% correct while Claude Sonnet 3.7 receives a score of 1%. This also points to a clear need for improvement in Finch’s performance, with 85% of bioinformatics tasks incorrect, and 52% of statistics tasks incorrect. The bioinformatics tasks required precise execution of multi-step pipelines to arrive at the correct answer. The statistics tasks typically involved single-step computations on clean tabular data. This indicates that at the time of submission, Finch was more likely to struggle on tasks that required execution of numerous steps in a row.

We agree with the reviewers that there have been substantial improvements in publicly available coding agents since the submission of Robin (as evidenced by the recent GPT-Rosalind model, linked above). However, the Nature editorial team has decided that it is not within scope to compare Robin to models or agents released after the original submission date of this manuscript, and has waived the requirement for us to do so. Therefore, we removed the comparisons to Claude Code and Biomni, both of which were released publicly after our manuscript was submitted. We hope that the included comparisons between Finch and its original underlying LLM Claude Sonnet 3.7 help to illustrate why a custom coding agent harness was necessary to include an autonomous data analysis component within the full Robin architecture at the time of submission.

Changes: Added new Finch evaluation to Results Section “Ablation studies and validation of the Robin architecture” (lines 224-237); Updated Supplementary Figure S18. Methods section “Finch evaluation” added. Added all question-answer pairs and evaluations to Supplementary Table S8.

2. The model comparisons are obsolete for a 2026 submission and must be updated. The paper uses OpenAI o4-mini as Robin’s backbone, Claude Sonnet 3.7 as the LLM judge, and compares against Claude Code (model version unspecified) and OpenAI Deep Research (June 2025). As of the date of this review, the frontier has advanced considerably (Claude Opus 4.5/4.6, GPT-5 class models, Gemini 3.1 Pro). Given that Claude Code already matched Finch with a mid-2025 model (point 1 above), it is reasonable to expect that current-generation general-purpose agents would equal or exceed Finch’s performance. Of course, the Robin backbone itself (o4-mini) is now several generations behind — the system’s own performance would likely improve substantially with updated models. For a paper submitted in 2026 to Nature, I consider it essential that the authors update the Finch comparison to include at least one current-generation coding agent (e.g., Claude Code with Opus 4.5 or later). The step-change in coding and analytical capabilities introduced by models such as Claude Opus 4.5 and GPT-5 class systems is well documented and represents a qualitative, not merely incremental, advance over the mid-2025 models used in this study. Given that Claude Code already matched Finch with a weaker model (Figure S18), the outcome with current-generation models bears directly on whether the specialized-agent

architecture remains justified. This comparison is straightforward to execute — it requires only providing the same dataset and prompt to an updated agent — and should not be deferred to a Discussion caveat.

Response: We appreciate the reviewer’s observation regarding the rapid advancement of frontier models (e.g., Claude Opus 4.5, GPT-5 class). We agree that the baseline capabilities of general-purpose agents have improved significantly since our mid-2025 submission. However, the editorial team has decided that it is not within scope to compare Robin to models released after the original submission date of this manuscript, and has waived the requirement to do so.

To address the reviewer’s valid point regarding the evolving landscape, we have added a discussion (line 281) of how current-generation frontier models interact with, and benefit from, specialized biological harnesses like Robin.

Changes: Discussion (Lines 281-285).

3. The biological validation is convincing as a proof-of-concept but should be framed accordingly. The RPE-SC experiments are a strong addition. The use of primary cells from an aged donor, fluorescently labeled rod outer segments (replacing beads), dose-response curves for both Y-27632 and ripasudil, immunocytochemistry validation of RPE markers, and the identification of KL001 as an additional hit collectively make a compelling in vitro case. The ABCA1 upregulation finding and its connection to AMD genetics via ApoE is an interesting mechanistic lead. However, the authors should be careful not to overstate these findings. This remains a single in vitro assay system (phagocytosis in cultured RPE cells), and as Reviewer 2 correctly noted, enhanced phagocytosis without concomitant increases in autophagy could theoretically worsen debris accumulation. The Discussion now appropriately mentions the need for in vivo validation, and I appreciate this addition.

Response: We thank the reviewer for this important nuance. We have gone through the manuscript and softened the claims of experimental findings (for example that Y-27632 may also enhance autophagy). We have also framed the identification of Y-27632, Ripasudil and KL001 as strong *in vitro* proof-of-concept rather than as promising therapeutics. We frame Robin as expediting the generation of in vitro data for candidate therapeutics molecules, rather than discovering novel therapeutics.

Changes: Abstract (lines 10-11) - removed reference to “validating” Ripasudil. Results (lines 159-162) - state that effect on autophagy of ROCKi requires further validation. Results (line 164) - changed ABCA1 to having a “potential” effect in dAMD. Amended Results to remove reference to expediting discovery of therapeutics (lines 169-171)

4. The time-on-task analysis (Table 1) is a useful addition but contains estimates that should be clearly flagged. Several entries in Table 1 are derived from published surveys (von Hippel et al. 2015 for hypothesis ideation; Sboner et al. 2011 and the Anaconda Data Science Report for bioinformatics setup) rather than measured directly in the context of this study. While these are reasonable estimates, they should be more prominently acknowledged as approximations to avoid giving the impression that the 200-fold speedup was empirically measured end-to-end.

Response: We have amended the Table 1 caption to more clearly state the the “Robin” estimates are directly measured while the human estimates are “derived from published

surveys". The publications are cited in the Table caption with symbols linking the references to the cited time-on-task estimates. Additionally, we amended the results section to state "using estimates from published surveys..."

Changes: Table 1 caption; Results (line 80).

Referee #1 (Remarks on code availability): My code review findings from the first round — specifically the rate-limiting errors that prevented full reproduction and Finch's restricted availability (not publicly released, accessible only upon request to FutureHouse) — do not appear to have been addressed in this revision.

Response: We apologize that the open-source version of Finch was not communicated in the previous letter or manuscript. The code for the version of Finch used in this manuscript is available here: <https://github.com/Future-House/finch>. This link has been added to the manuscript's Data and Code Availability section. This version of Finch can be installed in a Docker environment, and results from its data analysis can be manually returned to the Robin workflow (as the variable `experimental_insights`) for iterative therapeutic candidate generation. The user will not be subject to FutureHouse/Edison Scientific rate limits when using this open source version of Finch.

Changes: Code Availability statement updated

Reviewer 2 Response

Reviewer pt 9. Opaque reasoning in pairwise comparison to rank hypotheses: The inherent complexity of LLMs makes it difficult to interpret or understand the underlying logic used in generating ranked hypotheses from pairwise comparisons. The authors must clarify to what degree end users have flexibility in understanding or adjusting the hypothesis ranking criteria provided by Robin. In addition, the authors should clarify how their pairwise comparison methodology addresses potential systemic biases, given the limited description of the underlying statistical framework. In particular, prior studies have shown that pairwise comparisons using LLMs may inadvertently favor publications with a more assertive or authoritative tone over those grounded in greater scientific rigor. The authors should explain how the current approach mitigates such biases, or consider alternative comparison methods (i.e. would pointwise comparison make more sense?).

Reviewer reply to the authors response to point 9: It remains unclear whether alternative approaches for hypothesis ranking were considered. This is important to clarify as there is no clear consensus in the literature regarding optimal strategies for LLM-based relative rankings. Of particular concern is position bias within pairwise comparison frameworks, and multiple variants exist that incorporate aspects of pointwise comparison to improve robustness (e.g., Choo et al., 2025; Bouchard et al., 2025). Further detail on the authors' choice of methodology would therefore be valuable. Therefore changes should be made to the text.

Response: We appreciate the push for greater transparency regarding our methodological choices. We have added a new subsection to the Methods detailing our rationale for selecting the pairwise comparison framework over alternative strategies. Specifically, we discuss how we addressed position bias by utilizing a highly distributed tournament-style methodology paired with the Bradley-Terry-Luce (BTL) statistical model. Hypothesis ranking was calculated via LLM-adjudicated pairwise comparisons: either a complete round-robin tournament for sets of 25 or fewer hypotheses, or 300 randomly selected pairings for larger sets. Crucially, we did not rely on simple win/loss tallies, which are highly susceptible to positional bias. Instead, the BTL model aggregates this large volume of randomized pairwise outcomes to estimate global strength parameters. By mathematically diluting the influence of individual LLM comparison errors, this approach yields a robust relative ranking while forcing the model to make direct, biologically reasoned trade-offs.

Changes: Added detailed methodological justification to Methods Section (lines 304-308)

10. Robin's reliance on LLMs introduces sensitivity to prompt phrasing. This renders its candidate ranking system potentially variable and limits the reproducibility of hypothesis prioritization, reducing user confidence in the consistency and robustness of the system's output. For example, the experimental assays recommended by the robin full.ipynb and robin demo.ipynb scripts for dry AMD vary significantly due to changes in search criteria. The authors should provide user guidelines on recommended settings that effectively mitigate the influence of prompt sensitivity on generated results.

Reviewer Reply to the authors response to point 10: While stochasticity is an inherent feature of LLMs, the explanation of whether this variability meaningfully parallels

human reasoning within the Robin framework remains inadequate. Further prompt-based analyses that distinguishes variability driven by underlying evidence from that arising due to semantic sensitivity would improve user confidence in recommended experiments and hypotheses. Therefore changes should be made to the text.

Response: Response: We thank the reviewer for highlighting the need for additional clarification on how to prompt Robin to yield consistent results. We have updated the GitHub README.md (line 89) to include further prompting guidance. We note that Robin is designed to reduce prompt-based variability in the hypothesis generation steps by taking only the name of a disease as input from a user, without additional context or qualifiers. With respect to the variability of the hypothesis ranking system after the initial set of hypotheses have been generated, we find that the LLM judge exhibits an 88% intra-rater consistency on identical pairwise comparisons, significantly outperforming human experts (61%; see lines 313-318 and Figure S14B).

Changes: Added recommended user guidelines to the GitHub README.

Remarks on code availability

Reviewers comments: Difficulty in software implementation and virtual environment set-up: While the proposed use of a virtual environment is effective for managing Python-specific dependencies, it may still be susceptible to compatibility issues arising from version mismatches—particularly as newer versions of Python and its associated packages are released. These conflicts can introduce challenges in maintaining reproducibility and long-term stability of the computational environment. In the course of this review, multiple installation attempts using both PC and Mac systems were attempted, and different bugs/installation issues using one or the other were encountered. This may or may not be due to residual dependencies or interactions between the local host’s operating system and the virtual environment. To address this, the authors should consider a more fully self-contained solution such as Docker, which can provide a uniform runtime environment irrespective of the underlying operating system. Additionally, when setting api keys, a consistent error message was encountered when running the Robin configuration command. It was unable to successfully utilize the pai key added to a local .env file. Again, this may or may not be due to the local machine setup, but a more self-contained installation program would make installation smoother.

Reviewer reply to the authors response: The authors do not appear to have addressed this issue. Our prior comments regarding code reproducibility and implementation variability were not directly addressed, and the associated github repository does not appear to have been updated since the initial review. Clarification on plans for improving reproducibility, such as through containerization of the demo code, would be helpful. Therefore changes should be made to address these comments.

Referee #2 (Remarks on code availability): please see notes added to the end of the first review. the authors didnt respond to our prior comments so its not clear that they received them.

Response: We apologize for the installation difficulties and for missing this critical feedback in the previous round. We have overhauled the repository to include a fully containerized Docker implementation. This resolves the OS-specific dependency conflicts and provides a uniform runtime environment. We have also patched the API key configuration bug that was preventing the ‘.env’ file from loading properly.

237 **Changes:** GitHub repository updated with a comprehensive ‘Dockerfile’ and ‘docker-
238 compose.yml’; API key loading logic fixed; ‘README.md’ updated with containerized
239 installation instructions.